

## Candidate Athila recombination events

Total candidate elements: 161

intra-Athila (between ATHILA families): 128 (2 full-5)

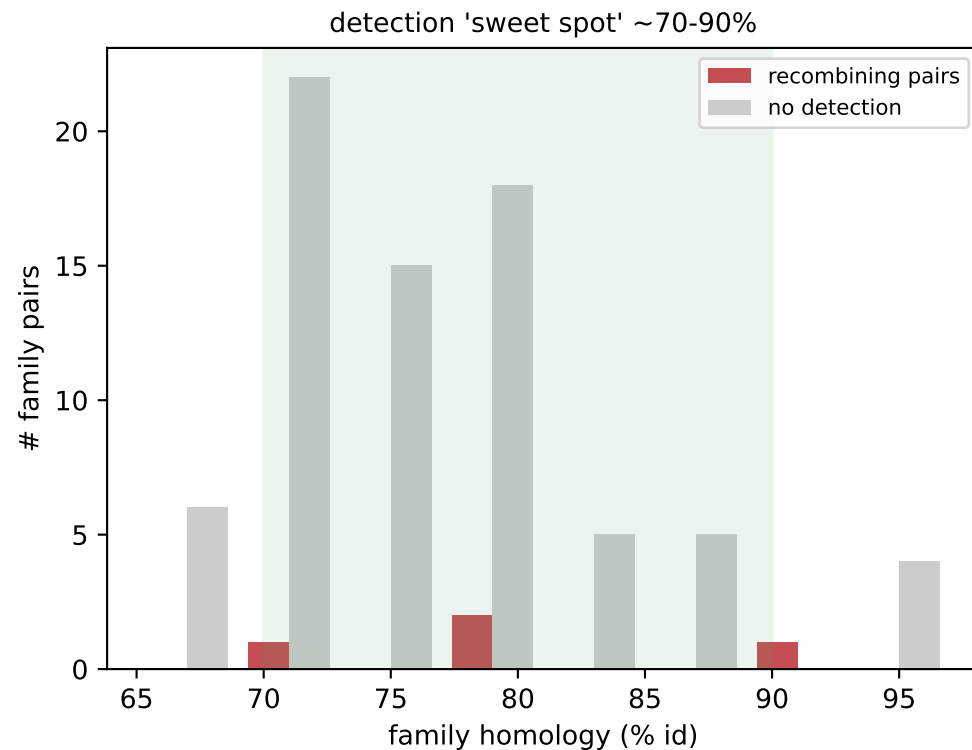
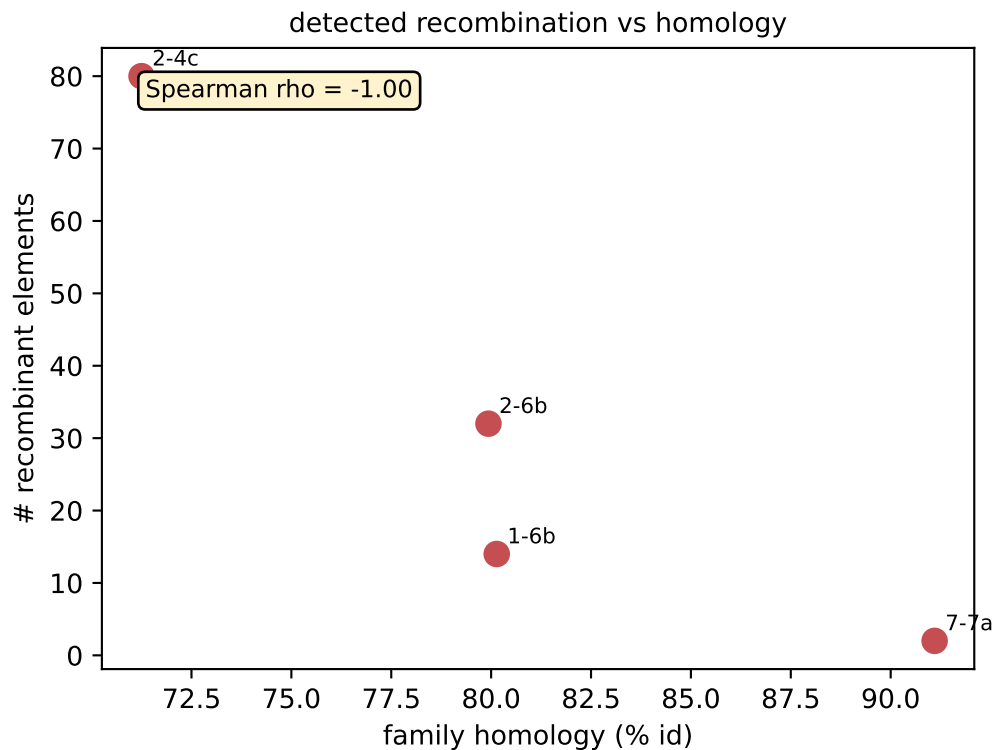
inter-clade (Athila x other LTR retro family): 33

Recombination sits at the element ENDS (GAG / INT) in every event type; the RT/RH core never swaps.

Confidence: intra-Athila = nucleotide per-domain family typing (margin  $\geq 1.3\times$ ); inter-clade = TEsorter per-domain clade call  
These are CANDIDATES (not yet Phase-6 nesting-filtered or Phase-7 breakpoint-confirmed).

| category     | partners         | swapped | # elem |
|--------------|------------------|---------|--------|
| intra_athila | ATHILA2,ATHILA4c | GAG     | 80     |
| intra_athila | ATHILA2,ATHILA6b | GAG     | 19     |
| inter_clade  | Athila+CRM       | CHDCR   | 16     |
| inter_clade  | Athila+Retand    | INT;aRH | 13     |
| intra_athila | ATHILA2,ATHILA6b | INT     | 13     |
| intra_athila | ATHILA1,ATHILA6b | INT     | 11     |
| intra_athila | ATHILA1,ATHILA6b | GAG     | 3      |
| inter_clade  | Athila+Ivana     | INT     | 2      |
| intra_athila | ATHILA7,ATHILA7a | GAG     | 2      |
| inter_clade  | Athila+CRM       | INT     | 1      |
| inter_clade  | Athila+Tork      | GAG     | 1      |

# Recombination bias vs family homology (Robin's question)

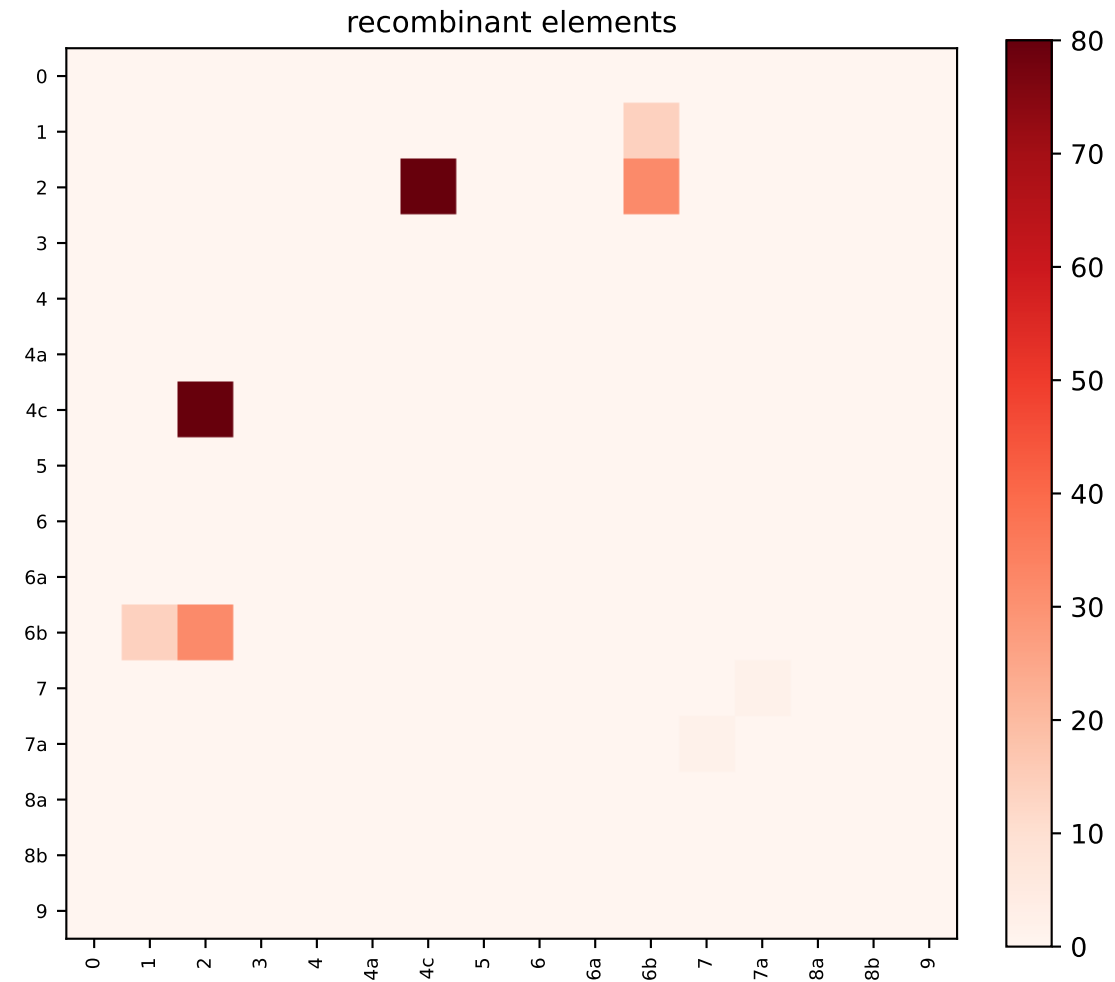
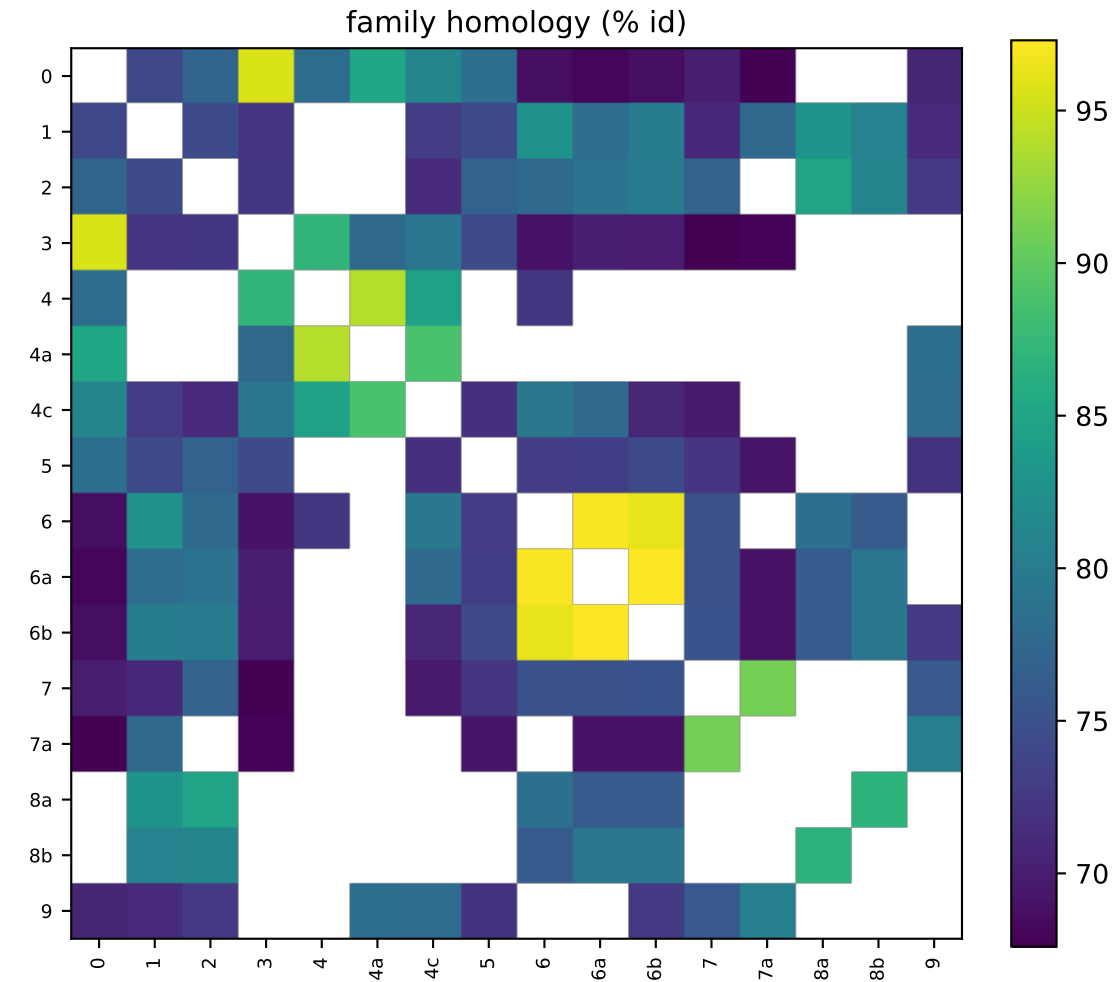


ANSWER (with the essential caveat):

- Raw detected recombination is **NEGATIVELY** correlated with homology ( $\rho = -1.0$  over the 4 detectable pairs), and the **MOST** homologous families (>93% id: ATHILA6a-6b, ATHILA0-3, ATHILA4-4a) show **ZERO** detections.
- This is largely a **DETECTION ARTEFACT**, not biology: when two families are near-identical, a swapped domain cannot be confidently assigned to one vs the other, so homologous-recombination between the most similar families is **INVISIBLE** to this method. Detection peaks at intermediate homology (~70-90%).
- Counts are also confounded by family **ABUNDANCE** (ATHILA2/4c/6b are common, so partner more often by chance).

To test the homology-bias hypothesis properly we need a method that does **NOT** rely on discriminating the swapped family: nucleotide breakpoint detection (GARD/RDP5) or diagnostic-SNP mosaic mapping, normalised by pairwise family abundance. The current data can only say recombination is readily detected between moderately-divergent Athila families.

## Family homology vs recombination matrix



## All candidate recombination events (rows 1-42 of 161)

| category     | accession                   | chrom | start    | architecture            | event                   | full5 |
|--------------|-----------------------------|-------|----------|-------------------------|-------------------------|-------|
| inter_clade  | ragtagCas-6_ColChr1to5      | Chr3  | 17625690 | GAG:Athila PROT:Athila  | INT=CRM                 | 0     |
| inter_clade  | ASS3_AT                     | Chr4  | 10134551 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Etna-2.ragtag_Ch1to5        | Chr4  | 10367893 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Etna-2_kang_ragtag_Ch1to5   | Chr4  | 10368859 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Rubezhnoe-1_fixed_Ch1to5    | Chr4  | 10470607 | GAG:Tork RT:Athila RH:A | GAG=Tork                | 0     |
| inter_clade  | ragtagMed-0_ColChr1to5      | Chr4  | 8305988  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | ragtagBELC-C-10_ColChr1to5  | Chr4  | 9092803  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | ragtagCvi-0_ColChr1to5      | Chr4  | 9092838  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | BELC-C-12.ragtag_Ch1to5     | Chr4  | 9099461  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | ragtag9762_ColChr1to5       | Chr4  | 9344328  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | _Anz-0_ColChr1to5ragtag     | Chr4  | 9493670  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Sf-2.ragtag_Ch1to5          | Chr4  | 9600249  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | AS530_AT                    | Chr4  | 9832191  | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Tibet-0.ragtag_Ch1to5       | Chr5  | 11312710 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Got-22.ragtag_Ch1to5        | Chr5  | 11946063 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | LI-OF-095.ragtag_Ch1to5     | Chr5  | 12487630 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Hs-0.ragtag_Ch1to5          | Chr5  | 12674576 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | _6184_Ch1to5                | Chr5  | 13239502 | GAG:Athila PROT:Athila  | CHDCR=CRM               | 0     |
| inter_clade  | Rab-R1.ragtag_Ch1to5        | Chr5  | 13594790 | GAG:Athila PROT:Athila  | INT=Ivana               | 0     |
| inter_clade  | 22005.ragtag_Ch1to5         | Chr5  | 13595757 | GAG:Athila PROT:Athila  | INT=Ivana               | 0     |
| inter_clade  | Kelsterbach-2.ragtag_Ch1to5 | Chr5  | 15393193 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | Dra-2.ragtag_Ch1to5         | Chr5  | 15739333 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | Est-0.ragtag_Ch1to5         | Chr5  | 15915508 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | _No-0_ColChr1to5ragtag      | Chr5  | 16146685 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | St-0_kang_ragtag_Ch1to5     | Chr5  | 16162089 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | 9994.ragtag_Ch1to5          | Chr5  | 16533472 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | ragtagMed-3_ColChr1to5      | Chr5  | 16874362 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | ragtagERR8705874_ColChr1to5 | Chr5  | 17027210 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | Nz-1.ragtag_Ch1to5          | Chr5  | 17135180 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | Stw-0_Ch1to5_2026           | Chr5  | 17311503 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | LL-0.ragtag_Ch1to5          | Chr5  | 17812374 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | ragtagBARA-C-5_ColChr1to5   | Chr5  | 17842294 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| inter_clade  | ragtagMr-0_ColChr1to5r      | Chr5  | 18674618 | GAG:Athila PROT:Athila  | INT=Retand;aRH=Retar    | 0     |
| intra_athila | Moh-0.ragtag_Ch1to5         | Chr1  | 18706603 | GAG:ATHILA1;INT:ATHILA1 | GAG > ATHILA1;ATHILA1 > | 0     |

## All candidate recombination events (rows 43-84 of 161)

| category     | accession                   | chrom | start    | architecture          | event                | full5 |
|--------------|-----------------------------|-------|----------|-----------------------|----------------------|-------|
| intra_athila | Sij-2.ragtag_Ch1to5         | Chr3  | 16472303 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | AH-7.ragtag_Ch1to5          | Chr3  | 16538225 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagEvs-12_ColChr1to5     | Chr3  | 16599339 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Kondara.ragtag_Ch1to5       | Chr3  | 16662274 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Sorbo.ragtag_Ch1to5         | Chr3  | 16668016 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagHum-4_ColChr1to5      | Chr3  | 16689894 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtag7143_ColChr1to5       | Chr3  | 16707844 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Kar-1.ragtag_Ch1to5         | Chr3  | 16752297 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Nov-2.ragtag_Ch1to5         | Chr3  | 16774580 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Sus-1.ragtag_Ch1to5         | Chr3  | 16813478 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Cdm-0.ragtag_Ch1to5         | Chr3  | 16888913 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagSan-9_ColChr1to5      | Chr3  | 17010882 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Sha.ragtag_Ch1to5           | Chr3  | 17031302 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagMr-0_ColChr1to5       | Chr3  | 17058671 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Elh-2.ragtag_Ch1to5         | Chr3  | 17059426 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | _Geg-14_ColChr1to5rag       | Chr3  | 17076446 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | LI-OF-095.ragtag_Ch1to5     | Chr3  | 17237601 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtag9852_Ch1to5           | Chr3  | 17299200 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Sf-2.ragtag_Ch1to5          | Chr3  | 17331321 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ws-4.ragtag_Ch1to5          | Chr3  | 17423380 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Per-1.ragtag_Ch1to5         | Chr3  | 17539770 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagCAMA-C-2_ColChr1to5   | Chr3  | 17560455 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagCAMA-C-9_ColChr1to5   | Chr3  | 17564115 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Belmonte-4-94.ragtag_Ch1to5 | Chr3  | 17623601 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtag9879_ColChr1to5       | Chr3  | 17655039 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagFERR-A-8_ColChr1to5   | Chr3  | 17665472 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagFERR-A-12_ColChr1to5  | Chr3  | 17676692 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | 9994.ragtag_Ch1to5          | Chr3  | 17724292 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Est-0.ragtag_Ch1to5         | Chr3  | 17736003 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | _6024_Ch1to5                | Chr3  | 17776000 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Hs-0.ragtag_Ch1to5          | Chr3  | 17811760 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtagCat-0_ColChr1to5      | Chr3  | 17837887 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | ragtag9336_ColChr1to5       | Chr3  | 17851800 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |
| intra_athila | Kac-1.ragtag_Ch1to5         | Chr3  | 17867002 | GAG:ATHILA2;RT:ATHILA | GAG->ATHILA2,ATHILA4 | 0     |

## All candidate recombination events (rows 85-126 of 161)

| category     | accession                   | chrom | start    | architecture           | event                 | full5 |
|--------------|-----------------------------|-------|----------|------------------------|-----------------------|-------|
| intra_athila | Dra-2.ragtag_Ch1to5         | Chr3  | 18017909 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtag9900_ColChr1to5       | Chr3  | 18030148 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagMet-6_ColChr1to5      | Chr3  | 18038631 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Bor-1.ragtag_Ch1to5         | Chr3  | 18061664 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ANGE-B-10.ragtag_Ch1to5     | Chr3  | 18061875 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagBANI-C-12_ColChr1to5  | Chr3  | 18071314 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagANGE-B-2_ColChr1to5   | Chr3  | 18073402 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagCas-6_ColChr1to5      | Chr3  | 18098240 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Elk-1.ragtag_Ch1to5         | Chr3  | 18098377 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagMERE-A-13_ColChr1to5  | Chr3  | 18116398 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Ishikawa.ragtag_Ch1to5      | Chr3  | 18204499 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Hiroshima.ragtag_Ch1to5     | Chr3  | 18213796 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Sah-0.ragtag_Ch1to5         | Chr3  | 18304635 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Abd-0.ragtag_Ch1to5         | Chr3  | 18366754 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagERR8705874_ColChr1to5 | Chr3  | 18428854 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | LL-0.ragtag_Ch1to5          | Chr3  | 18466328 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Qar-8a.ragtag_Ch1to5        | Chr3  | 18535407 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Yilong-0.ragtag_Ch1to5      | Chr3  | 18556230 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Co-4.ragtag_Ch1to5          | Chr3  | 18576738 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Tul-0.ragtag_Ch1to5         | Chr3  | 18593259 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagHum-2_ColChr1to5      | Chr3  | 18665765 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | RRS10.ragtag_Ch1to5         | Chr3  | 18725297 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Ket-10.ragtag_Ch1to5        | Chr3  | 18848402 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | 100621.Chr_scaffolds        | Chr3  | 18892645 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagMdc-14_ColChr1to5     | Chr3  | 18940581 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | 9806.ragtag_Ch1to5          | Chr3  | 18941716 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagHom-0_ColChr1to5      | Chr3  | 19133772 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagCas-0_ColChr1to5      | Chr3  | 19208025 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Tanz_merc.ragtag_Ch1to5     | Chr3  | 19545659 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | Tanz_wlod.ragtag_Ch1to5     | Chr3  | 19760968 | GAG:ATHILA2;RT:ATHILA  | GAG->ATHILA2,ATHILA4  | 0     |
| intra_athila | ragtagBELC-C-10_ColChr1to5  | Chr4  | 10008788 | GAG:ATHILA2;INT:ATHILA | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | BELC-C-12.ragtag_Ch1to5     | Chr4  | 10015446 | GAG:ATHILA2;INT:ATHILA | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagCvi-0_ColChr1to5      | Chr4  | 10121096 | GAG:ATHILA2;INT:ATHILA | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | Sf2.ragtag_Ch1to5           | Chr4  | 10508670 | GAG:ATHILA2;INT:ATHILA | GAG->ATHILA2,ATHILA6  | 0     |

## All candidate recombination events (rows 127-161 of 161)

| category     | accession               | chrom | start    | architecture          | event                 | full5 |
|--------------|-------------------------|-------|----------|-----------------------|-----------------------|-------|
| intra_athila | LL-0.ragtag_Ch1to5      | Chr4  | 8173043  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | Mammo-1.ragtag_Ch1t     | Chr4  | 8197597  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | Kar-1.ragtag_Ch1to5     | Chr4  | 8218560  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | _No-0_ColChr1to5ragtag  | Chr4  | 8277871  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagMr-0_ColChr1to5r  | Chr4  | 8292674  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | ragtagCas-0_ColChr1to5  | Chr4  | 8347855  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | ragtagBANI-C-12_ColCh   | Chr4  | 8509799  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | _Dog-4_ColChr1to5ragta  | Chr4  | 8613500  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | Got-22.ragtag_Ch1to5    | Chr4  | 8736316  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagMed-0_ColChr1to   | Chr4  | 8864452  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | _6184_Ch1to5            | Chr4  | 8940665  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | 100623.Chr_scaffolds    | Chr4  | 8951130  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | Meh-0.ragtag_Ch1to5     | Chr4  | 9131654  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagAlo-19_ColChr1to  | Chr4  | 9157877  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagMdc-14_ColChr1t   | Chr4  | 9212033  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | Pu2-23_ragtagChr1to5    | Chr4  | 9247597  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagBARC-A-12_ColCh   | Chr4  | 9347366  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagBARC-A-17_ColCh   | Chr4  | 9380767  | GAG:ATHILA2;INT:ATHIL | INT->ATHILA2,ATHILA6b | 0     |
| intra_athila | _6024_Ch1to5            | Chr4  | 9583005  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | ragtagMos-9_ColChr1to   | Chr4  | 9781497  | GAG:ATHILA2;INT:ATHIL | GAG->ATHILA2,ATHILA6  | 0     |
| intra_athila | Got-22.ragtag_Ch1to5    | Chr5  | 11832510 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Sus-1.ragtag_Ch1to5     | Chr5  | 12190663 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | ragtagBANI-C-12_ColCh   | Chr5  | 12231143 | GAG:ATHILA1;INT:ATHIL | GAG->ATHILA1,ATHILA6  | 0     |
| intra_athila | LI-OF-095.ragtag_Ch1t   | Chr5  | 12373227 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Hs-0.ragtag_Ch1to5      | Chr5  | 12559129 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | _Altai-5_ColChr1to5ragt | Chr5  | 12649958 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Meh-0.ragtag_Ch1to5     | Chr5  | 12729901 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Taz-0.ragtag_Ch1to5     | Chr5  | 12920487 | GAG:ATHILA1;INT:ATHIL | GAG->ATHILA1,ATHILA6  | 0     |
| intra_athila | Sorbo.ragtag_Ch1to5     | Chr5  | 12999141 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | _6184_Ch1to5            | Chr5  | 13133224 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Kondara.ragtag_Ch1to5   | Chr5  | 13427325 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Kar-1.ragtag_Ch1to5     | Chr5  | 13433921 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Kyr-1.ragtag_Ch1to5     | Chr5  | 13711584 | GAG:ATHILA1;INT:ATHIL | INT->ATHILA1,ATHILA6b | 0     |
| intra_athila | Toufl-1.ragtag_Ch1to5   | Chr5  | 18230639 | GAG:ATHILA7a;PT:ATHIL | GAG->ATHILA7,ATHILA7  | 1     |